Day 8 — JavaScript Fundamentals

Complete solutions to the tasks in the uploaded assignment.

How to run

Save the code in a file named **day8.js** and run it with Node.js, or paste each program into your browser console. Each example is independent unless noted otherwise.

1. Programs Using Conditions and Loops

1A. Check whether a number is positive, negative, or zero

```
let number = -5;

if (number > 0) {
    console.log("Positive number");
} else if (number < 0) {
    console.log("Negative number");
} else {
    console.log("Zero");
}

// Output: Negative number
```

1B. Print numbers from 1 to 10 using a loop

```
for (let i = 1; i <= 10; i++) {
    console.log(i);
}

// Output: 1 2 3 4 5 6 7 8 9 10 (each on a new line)
```

1C. Find the sum of numbers from 1 to 10

```
let sum = 0;

for (let i = 1; i <= 10; i++) {
    sum += i;
}

console.log("Sum:", sum);

// Output: Sum: 55
```

1D. Check whether a number is even or odd

```
let number = 7;

if (number % 2 === 0) {
    console.log("Even");
} else {
    console.log("Odd");
}

// Output: Odd
```

2. Functions for Basic Calculations

2A. Addition, subtraction, multiplication, and division

```
function add(a, b) {
    return a + b;
```

```
}

function subtract(a, b) {
    return a - b;
}

function multiply(a, b) {
    return a * b;
}

function divide(a, b) {
    if (b === 0) {
        return "Cannot divide by zero";
    }
    return a / b;
}

console.log(add(10, 5));       // 15
console.log(subtract(10, 5));  // 5
console.log(multiply(10, 5));  // 50
console.log(divide(10, 5));    // 2
console.log(divide(10, 0));    // Cannot divide by zero
```

2B. Find the square of a number

```
function square(number) {
    return number * number;
}

console.log(square(6));

// Output: 36
```

3. Array Operations

3A. Create an array and display its elements

```
let fruits = ["Apple", "Banana", "Mango"];

console.log(fruits);

// Output: [ 'Apple', 'Banana', 'Mango' ]
```

3B. Add and remove array elements

```
let fruits = ["Apple", "Banana"];

fruits.push("Mango");       // Add at the end
console.log(fruits);

fruits.pop();               // Remove the last element
console.log(fruits);

// Output:
// [ 'Apple', 'Banana', 'Mango' ]
// [ 'Apple', 'Banana' ]
```

3C. Find the sum of array numbers

```
let numbers = [10, 20, 30, 40];
let sum = 0;

for (let number of numbers) {
    sum += number;
}
```

```
console.log("Sum:", sum);

// Output: Sum: 100
```

3D. Find the largest number in an array

```
let numbers = [12, 45, 7, 89, 23];
let largest = numbers[0];

for (let number of numbers) {
    if (number > largest) {
        largest = number;
    }
}

console.log("Largest:", largest);

// Output: Largest: 89
```

4. Create and Access Object Properties

4A. Create an object and access its properties

```
let student = {
    name: "Asha",
    age: 21,
    course: "Computer Engineering"
};

console.log(student.name);
console.log(student.age);
console.log(student.course);

// Output:
// Asha
// 21
// Computer Engineering
```

4B. Add and update an object property

```
let student = {
    name: "Asha",
    age: 21
};

student.city = "Mumbai"; // Add a property
student.age = 22;        // Update a property

console.log(student);

// Output: { name: 'Asha', age: 22, city: 'Mumbai' }
```

5. Basic Problem-Solving Programs

5A. Calculate factorial of a number

```
let number = 5;
let factorial = 1;

for (let i = 1; i <= number; i++) {
    factorial *= i;
}

console.log("Factorial:", factorial);

// Output: Factorial: 120
```

5B. Reverse a string

```
let text = "JavaScript";
let reversed = "";

for (let i = text.length - 1; i >= 0; i--) {
    reversed += text[i];
}

console.log(reversed);

// Output: tpircSavaJ
```

5C. Check whether a word is a palindrome

```
let word = "madam";
let reversed = word.split("").reverse().join("");

if (word === reversed) {
    console.log("Palindrome");
} else {
    console.log("Not a palindrome");
}

// Output: Palindrome
```

5D. Print a multiplication table

```
let number = 5;

for (let i = 1; i <= 10; i++) {
    console.log(`${number} x ${i} = ${number * i}`);
}

// Output:
// 5 x 1 = 5
// 5 x 2 = 10
// ...
// 5 x 10 = 50
```

Quick revision

• **Variables:** use `let` for values that can change and `const` for bindings that should not be reassigned.

• **Conditions:** `if / else if / else` select which code runs.

• **Loops:** `for` and `for...of` repeat actions.

• **Functions:** reusable blocks of code; `return` sends a result back.

• **Arrays:** ordered lists; `push()` adds to the end and `pop()` removes the last item.

• **Objects:** store information as key-value properties, accessed with dot notation such as `student.name`.